Mode-Aligned Rademacher Probes and Curvature-Weighted Allocation: Threefold Fewer Hessian–Vector Products at Gradient-Variance Parity, and Why Not Fivefold

Anonymous authors

Abstract

Projected Hessian learning supervises the curvature of machine-learning interatomic potentials through Hessian–vector products (HVPs) along random probes, one per configuration in every minibatch. We ask how few HVPs per stochastic-gradient step preserve both the unbiasedness and the variance of the resulting gradient estimator. The answer depends on where its noise lives. Most of it comes from the probes rather than from minibatch sampling. For Gaussian probes, much of it is chi-square amplitude noise along the eigendirections of the curvature error, which Rademacher signs in an aligned basis remove exactly; the reference normal modes provide such a basis for free. The rest is heavy tailed across configurations and predictable before any HVP is computed. Our estimator, MODAL, combines mode-aligned Rademacher probes, a zero-HVP curvature score and Horvitz–Thompson-weighted systematic sampling of the configurations that receive a probe; its loss and gradient are provably unbiased. When per-configuration exact gradients are nearly orthogonal, as we verify, any single-probe allocation of this kind can reduce the probe count by at most $\kappa\,\gamma_{\mathrm{law}}\,\gamma_{\mathrm{alloc}}$ under a variance budget κ, where the two factors are measurable probe-law and allocation gains. On a reactive-chemistry benchmark, with every run trained to the published stopping rule, five paired seeds and a pre-registered protocol, MODAL uses $3.01\times$ fewer HVPs per step than our reimplementation of projected Hessian learning (131.5 versus 395.3). Energy, force and Hessian errors on the test, intrinsic-reaction-coordinate and normal-mode-sampled sets are statistically indistinguishable from the baseline in all nine cells. On these sets the total gradient variance is 1.54–1.76 times the baseline's averaged over training, exceeding twice the baseline only marginally at the earliest checkpoint; on the training set it is 1.90 times. A pre-registered fivefold variant shows the limit at work: its variance rises to 2.54–2.93 times the baseline's, as the measured gains predict.

1 Introduction

Second derivatives of the potential energy surface determine vibrational spectra, transition-state curvature and phonons, and machine-learning interatomic potentials (MLIPs) trained only on energies and forces can reproduce first-order quantities well while missing them [Loew et al., 2025, Rodriguez et al., 2025b]. Supervising the full Hessian is quadratic in the number of atoms. Projected curvature supervision avoids this by matching Hessian–vector products (HVPs) along random probe directions: the Hessian loss becomes a Hutchinson trace estimate whose cost is one HVP per probe [Hutchinson, 1990, Rodriguez et al., 2026, Koker et al., 2026, Cui et al., 2026]. Projected Hessian learning (PHL) showed that one fresh Gaussian probe per configuration per minibatch recovers the accuracy of full-Hessian training at near force cost [Rodriguez et al., 2026]. What has not been asked is whether

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one probe per configuration per step is the right operating point. Every HVP adds compute, and when reference HVPs come from quantum chemistry on the fly, every probe is also a pair of force evaluations. Yet each probe delivers only a noisy gradient. Choosing the number of probes is therefore a variance–cost trade-off in stochastic gradient estimation: SGD convergence depends on the second moment of the stochastic gradient [Bottou et al., 2018], so the question is how few HVPs per step can keep that second moment where PHL puts it, without biasing the objective.

We answer this question for unbiased estimators that probe a subset of the minibatch, and find that the answer is governed by where the noise of the Hessian-matching gradient lives. Three empirical facts organise the study. First, the per-step Hessian gradient of PHL is dominated by *probe* noise rather than by minibatch sampling: on the reactive-chemistry benchmark of Rodriguez et al. [2026], the variance due to the probes is 2.2–5.4 times the between-minibatch variance of the exact gradient. Second, a large part of that probe noise is an artefact of the Gaussian law: the gradient is a vector of quadratic forms in the probe whose coefficient matrices are anticommutators of the curvature error with Hessian sensitivities, and a Gaussian probe pays an amplitude (chi-square) fluctuation along every direction that Rademacher signs remove exactly (Proposition 2). The reference normal modes are a free basis in which that removal is large. Third, the remaining noise is concentrated in a heavy-tailed minority of configurations (small molecules, stiff curvature, poorly fitted forces), which can be identified before any HVP is computed.

These facts give MODAL (mode-aligned probes with allocation): Rademacher probes in each molecule's reference normal-mode basis, assigned to a fixed fraction of the minibatch by randomised systematic probability-proportional-to-size sampling with a zero-HVP curvature score, and reweighted by inverse inclusion probabilities so that the Hessian loss and its gradient remain exactly unbiased. They also give a limit. For any single-probe unbiased allocation, the largest probe reduction compatible with a variance budget κ (relative to PHL) is $r^\star = \kappa\,\gamma_{\mathrm{law}}\,\gamma_{\mathrm{alloc}}$, the product of a probe-law gain and an allocation gain (Proposition 4). Measured on this benchmark, those gains cap the variance-preserving reduction at roughly three- to fourfold. Among the evaluation sets, the intrinsic-reaction-coordinate geometries are the binding case, and the training set is tighter still.

Our contributions are:

- A variance analysis of one-probe Hessian-matching gradients: an exact decomposition into amplitude and mixing noise for any orthonormal probe basis (Proposition 2), the variance-optimal Horvitz–Thompson allocation of probes across configurations (Proposition 3), and the probe-reduction limit $r^\star = \kappa\gamma_{\mathrm{law}}\gamma_{\mathrm{alloc}}$ (Proposition 4).
- MODAL, an unbiased estimator that uses $3.01\times$ fewer HVPs per SGD step than PHL (131.5 versus 395.3 time-averaged) with $O(B)$ classical overhead per step.
- A pre-registered evaluation on Test, IRC and NMS with five paired seeds, trained to the published stopping rule (no reduced schedule): accuracy parity with reimplemented PHL on all nine energy/force/Hessian cells under the paired t-test, and total gradient variance 1.56, 1.76 and 1.54 times PHL's, averaged over training. On the evaluation sets the variance exceeds $2\times$ only at the earliest audited checkpoint (epoch 5), by a borderline margin.
- A test of the limit: the pre-registered fivefold variant keeps accuracy parity ($9/9$ cells) but carries 2.54–$2.93\times$ PHL's gradient variance, as the measured γ factors predict. Ablations attribute the saving to the probe law and the allocation separately, and a mechanism, failure and cost analysis explains where the method helps and where it cannot.

2 Related work

Curvature supervision for interatomic potentials. Training on energies and forces is standard for machine-learning interatomic potentials [Smith et al., 2017, Lubbers et al., 2018], but second-derivative observables such as vibrational frequencies, transition-state curvature and phonons can remain inaccurate [Loew et al., 2025, Yuan et al., 2024]. Explicit Hessian losses improve them [Rodriguez et al., 2025b, Williams et al., 2025, Amin et al., 2025] at a cost quadratic in the number of atoms. Projected supervision through Hessian–vector products replaces the full Hessian by random projections: one-column (one-hot) probing samples Hessian columns [Cui et al., 2026, Koker et al., 2026, Amin et al., 2025], projected Hessian learning (PHL) uses Hutchinson probes and shows that per-minibatch randomisation recovers full-Hessian accuracy at near force cost [Rodriguez et al.,

2026], and related stochastic projections appear in Hessian-informed training protocols [Yin et al., 2026] and in coarse-grained Hessian matching [Murdeshwar et al., 2026]. Architectures that predict Hessians directly [Burger et al., 2025] sidestep automatic differentiation altogether. All projected methods we are aware of spend one probe per configuration per step and treat the probe law as given; none analyses how the probe budget should be allocated across configurations or how the probe law interacts with the structure of the curvature error.

Stochastic trace estimation. Hutchinson's estimator [Hutchinson, 1990, Girard, 1989] and its analyses [Avron and Toledo, 2011, Roosta-Khorasani and Ascher, 2015] establish that Rademacher probes minimise the variance of $v^\top Av$ among i.i.d. laws, removing the diagonal contribution that Gaussian probes carry. Variance reduction for *many* probes on *one* matrix exploits low-rank structure (Hutch++ [Meyer et al., 2021], XTrace [Epperly et al., 2024], randomised sketching [Halko et al., 2011]) or sparsity/colouring (probing [Tang and Saad, 2012, Bekas et al., 2007]). Hutchinson estimators are also used inside training objectives—sliced score matching [Song et al., 2019], continuous normalising flows [Grathwohl et al., 2019], physics-informed networks [Hu et al., 2024]—where Hutch++ has recently been adapted to reduce training variance [Liu et al., 2025]. Our setting differs in two ways that change the answer: the estimated quantity is the *parameter gradient* of a trace (a vector-valued quadratic form whose coefficient matrices are anticommutators of the curvature error with Hessian sensitivities), and the budget is far below one probe per matrix, so the question is which of many small matrices to probe at all. Proposition 2 is the gradient analogue of the classical Rademacher result in an arbitrary orthonormal basis; using the reference normal modes [Wilson et al., 1955] as that basis is, to our knowledge, new.

Variance reduction and importance sampling in SGD. Importance sampling of training examples [Needell et al., 2014, Zhao and Zhang, 2015, Alain et al., 2015, Stich et al., 2017, Katharopoulos and Fleuret, 2018, Johnson and Guestrin, 2018] and of minibatches [Csiba and Richtárik, 2018, Gower et al., 2019] chooses sampling probabilities proportional to per-example gradient norms or their bounds; Horvitz–Thompson reweighting with defensive mixtures has recently been used to skip low-loss examples in the backward pass [Bonbhel, 2026]. Control-variate methods (SVRG, SAGA, STORM) [Johnson and Zhang, 2013, Defazio et al., 2014, Cutkosky and Orabona, 2019] reuse stored gradients or paired evaluations; their benefit for deep networks is limited by fast parameter drift [Defazio and Bottou, 2019], and in our setting they would require re-evaluating probes at two parameter values, doubling the HVP count. We borrow the survey-sampling toolkit—Horvitz–Thompson estimation [Horvitz and Thompson, 1952], systematic PPS designs [Madow, 1949], Neyman-type allocation [Neyman, 1934, Särndal et al., 1992] and defensive mixtures [Hesterberg, 1995]—but the sampling unit is a probe on a configuration rather than an example, the noise being allocated is mostly *internal* probe noise rather than between-example variation, and the allocation score is derived from the structure of the Hessian-matching gradient and computed before any HVP. The limit in Proposition 4 makes explicit, for this class, how much of a probe saving can come from the probe law and how much from heterogeneity across configurations.

3 Probe-efficient estimators of the Hessian-matching gradient

3.1 Setting

A minibatch $\mathcal{B}$ contains B molecular configurations. Configuration m has N_m atoms and $n_m = 3N_m$ Cartesian degrees of freedom; $\tilde{H}_m(\theta) \in \mathbb{R}^{n_m \times n_m}$ is the Hessian of the learned energy, H_m the reference Hessian and $E_m = \tilde{H}_m - H_m$ the (symmetric) curvature error. The Hessian term of the training objective is

$$\mathcal{L}_H(\theta; \mathcal{B}) = \frac{1}{B} \sum_{m \in \mathcal{B}} L_m, \qquad L_m = \frac{\|E_m\|_F^2}{n_m^2}, \tag{1}$$

and it enters $\mathcal{L} = \mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \mathcal{L}_H$ with $\lambda_F = 0.30$, $\lambda_H = 0.09$. A probe $v \in \mathbb{R}^{n_m}$ with $\mathbb{E}[vv^\top] = I$ yields the one-probe estimate

$$\ell_m(v) = \frac{\|\tilde{H}_m v - H_m v\|^2}{n_m^2} = v^\top A_m v, \qquad A_m = E_m^2 / n_m^2, \tag{2}$$

whose model part $\tilde{H}_m v$ is one Hessian–vector product (HVP) obtained by reverse-over-reverse differentiation [Pearlmutter, 1994], and whose reference part $H_m v$ is a data lookup. Projected

Hessian learning (PHL) [Rodriguez et al., 2026] draws one Gaussian probe for every molecule of every minibatch, i.e. B HVPs per step. Its gradient with respect to a parameter θ_p is a quadratic form in the probe,

$$g_{m,p}(v) = \frac{\partial \ell_m(v)}{\partial \theta_p} = v^\top S_{m,p}\, v, \qquad S_{m,p} = \frac{E_m D_{m,p} + D_{m,p} E_m}{n_m^2}, \quad D_{m,p} = \frac{\partial \tilde{H}_m}{\partial \theta_p}, \tag{3}$$

with mean $\bar{g}_m = \nabla_\theta L_m$ and second moment $s_m^2(\mathcal{P}) = \mathbb{E}\|g_m(v)\|^2$ under probe law $\mathcal{P}$. Every statement below concerns the θ-gradient of $\mathcal{L}_H$, which is what the optimiser consumes; all variances are traces of covariance matrices over the full parameter vector.

3.2 Estimator family and unbiasedness

We consider estimators that (i) select a random subset $S \subseteq \mathcal{B}$ with inclusion probabilities $\pi_m = \Pr(m \in S)$, (ii) draw one probe v_m from a law $\mathcal{P}_m$ for every selected molecule, independently of S, and (iii) reweight by the inverse inclusion probability [Horvitz and Thompson, 1952]:

$$\hat{\mathcal{L}}_H = \frac{1}{B} \sum_{m \in S} \frac{\ell_m(v_m)}{\pi_m}. \tag{4}$$

The number of HVPs in the step is $|S|$; PHL is the member with $S = \mathcal{B}$, $\pi_m \equiv 1$ and Gaussian probes.

Proposition 1 (Unbiasedness)**.** *If $\pi_m > 0$ for all m, the π_m are fixed before S and the probes are drawn (they may depend on θ, the data and past information, but not on the current probes), and every probe law satisfies $\mathbb{E}[v_m v_m^\top] = I$, then $\mathbb{E}[\hat{\mathcal{L}}_H] = \mathcal{L}_H$ and $\mathbb{E}[\nabla_\theta \hat{\mathcal{L}}_H] = \nabla_\theta \mathcal{L}_H$, with π_m treated as constants in the backward pass.*

Proof. For a fixed symmetric A, $\mathbb{E}[v^\top A v] = \operatorname{tr}(A\,\mathbb{E}[vv^\top]) = \operatorname{tr} A$, so $\mathbb{E}[\ell_m(v_m)] = \operatorname{tr} A_m = L_m$. By independence of S and the probes, $\mathbb{E}[\hat{\mathcal{L}}_H] = \frac{1}{B}\sum_m \mathbb{E}[\mathbf{1}\{m \in S\}]\,\mathbb{E}[\ell_m(v_m)]/\pi_m = \frac{1}{B}\sum_m L_m$. Since $\ell_m(v)$ is a polynomial in v whose coefficients are differentiable in θ and the probe laws have finite fourth moments, differentiation and expectation commute, which gives the gradient statement. □

Two probe laws are used. The Gaussian law $v \sim \mathcal{N}(0, I)$ is PHL's. The *mode-aligned Rademacher* law is $v = U_m \sigma$, where $H_m = U_m \Lambda_m U_m^\top$ is the eigendecomposition of the reference Hessian (its columns are the Cartesian normal modes) and $\sigma \in \{\pm 1\}^{n_m}$ has independent uniform signs; $\mathbb{E}[vv^\top] = U_m \mathbb{E}[\sigma\sigma^\top] U_m^\top = U_m U_m^\top = I$. Unbiasedness holds for *every* orthogonal U_m; the choice of basis only affects variance. For the subset designs we use randomised systematic probability-proportional-to-size sampling [Madow, 1949]: units are put in random order, their cumulative inclusion probabilities tile $[0, P)$, and the units hit by the points $u, u+1, \ldots, u+P-1$ ($u \sim U[0,1)$) are selected. This design selects exactly $P = \sum_m \pi_m$ molecules (so the per-step probe count is deterministic) and satisfies $\Pr(m \in S) = \pi_m$ exactly, hence Proposition 1 applies.

3.3 Where the probe noise lives: amplitude versus mixing

Proposition 2 (Probe-law variance)**.** *For any molecule m and any orthogonal U, write $M'_p = U^\top S_{m,p} U$. Then*

$$\operatorname{Var}_{\text{Gauss}}[g_m] = 2 \sum_p \sum_{k,l} (M'_{p,kl})^2 = 2 \sum_p \|S_{m,p}\|_F^2, \tag{5}$$

$$\operatorname{Var}_{U\text{-}Rad}[g_m] = 2 \sum_p \sum_{k \neq l} (M'_{p,kl})^2, \tag{6}$$

so that $\operatorname{Var}_{\text{Gauss}} - \operatorname{Var}_{U\text{-}Rad} = 2\sum_p \sum_k (M'_{p,kk})^2 \geq 0$ *for every basis U.*

Proof. For a symmetric M and $v = U\sigma$, $v^\top M v - \operatorname{tr} M = \sum_{k \neq l} \sigma_k \sigma_l M'_{kl}$ because $\sigma_k^2 = 1$; the products $\sigma_k \sigma_l$ ($k < l$) are uncorrelated with unit variance, giving $2\sum_{k \neq l} M'^2_{kl}$. For Gaussian z the law is rotation invariant and $v^\top M v - \operatorname{tr} M = \sum_k (z_k^2 - 1) M'_{kk} + \sum_{k \neq l} z_k z_l M'_{kl}$ with $\operatorname{Var}(z_k^2 - 1) = 2$, giving $2\|M\|_F^2$. Summing over parameters p yields the claim. □

The Gaussian probe therefore pays an *amplitude* (chi-square) noise $\sum_k (z_k^2 - 1) M'_{p,kk}$ along every direction of every basis, on top of the basis-dependent *mixing* noise $\sum_{k \neq l}$. Rademacher signs remove the amplitude term exactly, and the gain is largest in a basis in which the matrices $S_{m,p} = \{E_m, D_{m,p}\}/n_m^2$ are close to diagonal. The ideal basis (that of E_m) is unknown without n_m HVPs, but the reference normal modes are free and are the natural candidate: when the learned potential reproduces the mode shapes and errs mainly on the mode curvatures, E_m is nearly diagonal in U_m. In the limit of a purely spectral error (E_m commuting with H_m), the mode-aligned estimate of the *loss* is exact, $\ell_m(U_m \sigma) = \sum_k (U_m^\top A_m U_m)_{kk} = L_m$ for every sign pattern, whereas the Gaussian estimate keeps variance $2\|A_m\|_F^2 > 0$. Cartesian Rademacher probes ($U = I$) are also never worse than Gaussian ones, but the Cartesian basis is not adapted to the error, and translations are spread over all coordinates. Note that both the translational modes (exactly) and, at stationary geometries, the rotational modes (approximately) lie in the null space of E_m and of every $D_{m,p}$, so they contribute nothing to the mode-aligned variance.

3.4 Which molecules to probe: variance-optimal allocation

Proposition 3 (Allocation)**.** *Under Poisson sampling (independent inclusions) and one probe per selected molecule,*

$$\mathrm{Var}\left[\nabla_\theta \hat{\mathcal{L}}_H \,\middle|\, \mathcal{B}\right] = \frac{1}{B^2} \sum_{m \in \mathcal{B}} \left(\frac{s_m^2}{\pi_m} - \|\bar{g}_m\|^2 \right), \tag{7}$$

and among designs with $\sum_m \pi_m = P$ and $0 < \pi_m \leq 1$ the variance is minimised by $\pi_m^\star = \min(1, c\, s_m)$, with c fixed by the budget.

Proof. Write $\nabla_\theta \hat{\mathcal{L}}_H = \frac{1}{B} \sum_m I_m g_m(v_m)/\pi_m$ with independent $I_m \sim \mathrm{Bernoulli}(\pi_m)$. The summands are independent with $\mathbb{E}\|I_m g_m/\pi_m\|^2 = s_m^2/\pi_m$ and mean $\bar{g}_m$, which gives (7). The objective $\sum_m s_m^2/\pi_m$ is convex on the box; the KKT conditions give $s_m^2/\pi_m^2 = \mu$ for interior coordinates, i.e. $\pi_m \propto s_m$, and $\pi_m = 1$ where this would exceed one. □

Systematic sampling adds the joint-inclusion term $B^{-2} \sum_{m \neq m'} (\pi_{mm'}/(\pi_m \pi_{m'}) - 1) \langle \bar{g}_m, \bar{g}_{m'} \rangle$, which vanishes when the exact per-molecule gradients are orthogonal; we measured them to be nearly so (Section 6.1), and in any case the variance audit measures the realised variance of the systematic design directly. The second moments s_m are not observable before probing, so the allocation needs a surrogate available at zero HVP cost. From (3), s_m^2 scales like $\|E_m\|_F^2 \, \|D_m\|^2/n_m^4$, i.e. like L_m/n_m^2 times a sensitivity factor, and on pilot data (five checkpoints, four sets, 600 molecules each) s_m^2 indeed has a Spearman correlation of 0.80–0.89 with L_m and of -0.73 to -0.49 with N_m. L_m itself is unknown before probing, so we replace $L_m^{1/2}$ by the geometric mean of two quantities that are available before any HVP: the reference curvature scale $h_m = \|H_m\|_F^2/n_m^2$ (absolute curvature errors grow with curvature) and the current force residual $f_m = \|\tilde{F}_m - F_m\|^2/n_m$, which is computed anyway for the force loss. The *allocation score* and inclusion probabilities are

$$q_m = \frac{(h_m f_m)^{1/4}}{N_m}, \qquad \pi_m = \min\left\{1,\ (1-\alpha)\, P \frac{q_m}{\sum_{m'} q_{m'}} + \alpha \frac{P}{B}\right\} \text{(water-filled to } \textstyle\sum_m \pi_m = P), \tag{8}$$

where the defensive mixture with the uniform design [Hesterberg, 1995] bounds every Horvitz–Thompson weight by $1/\pi_m \leq B/(\alpha P)$ and keeps Proposition 1 valid even when the score is badly wrong. The exponents $(\frac{1}{4}, \frac{1}{4}, -1)$ were fixed from this scaling argument; a grid search on the pilot diagnostics (Section 4) confirmed they are within 0.3% of the best grid point in worst-case predicted variance and they were not tuned further.

3.5 How far can the probe count fall? A variance-budget limit

Let V_{data} be the between-minibatch variance of the exact minibatch gradient $\nabla_\theta \mathcal{L}_H(\mathcal{B})$, a quantity common to all unbiased estimators. The variance that enters SGD convergence bounds is the total mean-squared error of the stochastic gradient around the full-data gradient,

$$V_{\mathrm{total}} = \mathbb{E}\left\|\nabla_\theta \hat{\mathcal{L}}_H - \nabla_\theta \mathcal{L}_H(\mathcal{D})\right\|^2 = V_{\mathrm{data}} + \mathbb{E}_{\mathcal{B}}\, \mathrm{Var}\left[\nabla_\theta \hat{\mathcal{L}}_H \,\middle|\, \mathcal{B}\right]. \tag{9}$$

Proposition 4 (Probe-reduction limit)**.** *Let $r = B/P$ be the probe-reduction factor of a single-probe estimator with law $\mathcal{P}$ and the optimal allocation of Proposition 3 (no capping active), and let $\overline{x}$ denote an average over molecules. If per-molecule exact gradients are orthogonal, so that $B\,V_{\mathrm{data}} = \overline{\|\bar{g}\|^2}$, then its total variance relative to PHL is*

$$\frac{V_{\mathrm{total}}(r)}{V_{\mathrm{total}}^{\mathrm{PHL}}} = r\,\frac{\overline{\big(s(\mathcal{P})\big)}^{\,2}}{\overline{s^2(\mathrm{Gauss})}} = \frac{r}{\gamma_{\mathrm{law}}\,\gamma_{\mathrm{alloc}}}, \qquad \gamma_{\mathrm{law}} = \frac{\overline{s^2(\mathrm{Gauss})}}{\overline{s^2(\mathcal{P})}}, \quad \gamma_{\mathrm{alloc}} = \frac{\overline{s^2(\mathcal{P})}}{\overline{\big(s(\mathcal{P})\big)}^{\,2}} \geq 1, \tag{10}$$

so the largest reduction compatible with a variance budget κ (relative to PHL) is $r^\star(\kappa) = \kappa\,\gamma_{\mathrm{law}}\,\gamma_{\mathrm{alloc}}$.

Proof. By (7) with $\pi_m = P s_m / \sum s$, the conditional variance is $B^{-1}(r\,\overline{s}^2 - \overline{\|\bar{g}\|^2})$; PHL's is $B^{-1}(\overline{s_{\mathrm{G}}^2} - \overline{\|\bar{g}\|^2})$. Adding $V_{\mathrm{data}} = B^{-1}\overline{\|\bar{g}\|^2}$ to both gives the ratio. □

Proposition 4 splits any single-probe saving into a *probe-law* gain γ_{law} (how much less noisy one probe is, including the irreducible signal part $\|\bar{g}_m\|^2$) and an *allocation* gain $\gamma_{\mathrm{alloc}} = 1 + \mathrm{CV}^2(s)$ (how heterogeneous the molecules are). Uniform subsampling with Gaussian probes has $\gamma_{\mathrm{law}} = \gamma_{\mathrm{alloc}} = 1$, so at a variance budget of $\kappa = 2$ it can cut probes only by $r^\star = 2$; reaching $r = 5$ at $\kappa = 2$ requires $\gamma_{\mathrm{law}}\gamma_{\mathrm{alloc}} \geq 2.5$. With a practical score q, γ_{alloc} is replaced by the achieved gain $\overline{s^2}/\overline{s^2\,(P/B)/\pi_q}$. Section 6 measures all of these quantities.

3.6 The estimator used in this work

MODAL (mode-aligned probes with allocation) combines the three ingredients. At every SGD step, after the forward pass and force computation for the whole minibatch: (1) compute q_m from (8) for all B molecules (h_m is precomputed, f_m comes from the force loss); (2) set $P = \max\{1, \mathrm{round}(\phi B)\}$ with probe fraction $\phi = 1/3$ (so $P = 133$ for $B = 400$ and $P = 23$ for the final minibatch of 69 molecules), $\alpha = 0.2$, and form π_m; (3) draw S by randomised systematic PPS sampling; (4) for $m \in S$ draw $v_m = U_m \sigma_m$, evaluate the HVPs on a graph that contains only the selected molecules, and add $\lambda_H \hat{\mathcal{L}}_H$ from (4) to the energy and force losses of the full minibatch; (5) take the AdamW/SGD step. Everything else—architecture, data, splits, minibatches, optimisers, schedule, loss weights and normalisations—is PHL's. A variant with $\phi = 1/5$ is pre-registered as a probe of the limit. The bases U_m are computed once per molecule (one dense $n_m \times n_m$ eigendecomposition, $n_m \leq 69$); per step, the classical overhead is one score evaluation per molecule, one random permutation and cumulative sum, and one $n_m \times n_m$ matrix–vector product per selected molecule.

4 Experimental setup

Data and splits. We use the ωB97X-D/6-31G(d) OpenREACT-CHON-EFH data [Rodriguez et al., 2025a] exactly as in Rodriguez et al. [2026]: reactant/transition-state/product geometries (RTP; 35,087 configurations) are split 80/10/10 into training (28,069), validation (3,508) and test (3,510) sets, and the intrinsic-reaction-coordinate set (IRC; 34,248 configurations along 574 paths) and the normal-mode-sampled set (NMS; 62,527 far-from-equilibrium configurations) are used only for evaluation. Molecules have 4–23 atoms (median 14). The split reproduces the authors' notebook procedure (conformation list in file order, self-atomic energies fitted by least squares on RTP, Python `random.shuffle` after seeding with the authors' seed 7289038); because the notebook interleaves logging initialisation with the shuffle, our permutation follows the published procedure but may differ from the authors' exact one. The six supplied HDF5 files are three byte-identical pairs; we use one of each.

Model and training protocol (held fixed). All runs use the authors' ANI-style architecture and training cell [Smith et al., 2017, Gao et al., 2020, Rodriguez et al., 2026], which is the architecture of the published PHL experiments (the authors also distribute a HIP-NN implementation [Lubbers et al., 2018] in a separate package, which we do not use): AEV featurisation with the notebook's radial/angular parameters, one $384 \to 256 \to 64 \to 256 \to 1$ CELU network per element, float64 arithmetic, AdamW (weights, lr 10^{-3}, weight decay 10^{-2}) and SGD (biases, lr 10^{-3}) [Loshchilov and Hutter, 2019], ReduceLROnPlateau (factor 0.5, patience 100) on the validation energy RMSE,

early stopping when the learning rate falls below 10^{-5} (maximum 5000 epochs), loss weights $\lambda_F = 0.30$, $\lambda_H = 0.09$ and the notebook's per-molecule normalisations. As in the notebook, the shuffled training set is collated once into 71 fixed minibatches (70×400 and one of 69) that are visited in the same order every epoch; to obtain seed-to-seed variation with a fixed split, the run seed permutes the training list before collation and seeds the initialisation, the probes and the selections. The best-validation-energy checkpoint is evaluated. The paper's supplementary information lists a learning rate of 10^{-4} whereas the notebook uses 10^{-3}; we follow the notebook, which is the designated reference implementation. Hessian errors are always computed against the exact model Hessian ($3N$ HVPs per molecule), with the notebook's RMSE conventions (energy per molecule; force per component, averaged per molecule; Hessian per element, averaged per molecule).

Training schedule and compute. Every run trains to the paper's own stopping rule; there is no reduced schedule, no data subsampling and no change of batch size. With the notebook's learning rate the rule stops runs after 1462–2925 epochs, longer than the ∼700–1000 epochs reported in the paper (Section 6.3 discusses this deviation). Seeds $\{0, 1, 2, 3, 4\}$ are used for every configuration, so all comparisons are paired by seed (same minibatches and initialisation). Runs were executed on a node with eight NVIDIA H200 GPUs, with up to four runs sharing a GPU through CUDA MPS; training was made resumable (probe and selection draws are seeded by run seed and global step, so a resumed run draws exactly the probes it would have drawn). Because shared-GPU epoch times are not comparable across configurations, the wall-clock column reports a separate single-process benchmark on an otherwise idle H200 (median of 20 epochs after 3 warm-up epochs).

Configurations. *PHL (paper as reported)* denotes the published numbers (Fig. 3a and Fig. 4 of Rodriguez et al. [2026], randomised probes). *PHL (reimplemented)* is the unmodified estimator run with the authors' code path (one Gaussian probe per molecule per step, all HVPs on the shared minibatch graph). *ModAL* is the estimator of Section 3.6 with probe fraction $1/3$ (main configuration) or $1/5$ (pre-registered limit probe). Ablations isolate each ingredient at the same seeds: mode-aligned probes with all molecules probed (same law, same count as PHL—the control required to separate a zero-cost probe-law improvement from a probe-count saving); uniform Horvitz–Thompson subsampling with Gaussian probes at $1/3$ and $1/5$ (no allocation, no new law); uniform subsampling with mode-aligned probes at $1/3$ (law only); curvature-weighted allocation with Gaussian probes at $1/3$ (allocation only). We also train one-column probing with all molecules (the main published alternative [Cui et al., 2026, Koker et al., 2026]) and energy–force training without a Hessian term.

Pre-registered evaluation. The protocol below was written before any final-seed run was launched (the document is part of the released logs). *Probe count*: the number of molecule-HVPs per SGD step is logged for every step; the reported value is its time average over the whole run (PHL: 395.3, because the last minibatch has 69 molecules). *Accuracy parity*: for each of the nine (Test, IRC, NMS)×(E, F, H) cells we apply a two-sided paired t-test across the five seeds (route (a)); a cell is at parity if $p > 0.05$. As secondary information only, we report a paired TOST equivalence test [Schuirmann, 1987] with margin $\pm 10\%$ of the reimplemented-PHL mean; route (a) does not by itself establish equivalence and we do not claim equivalence where TOST fails. *Variance parity*: at checkpoints $\{5, 25, 100, 200, 400, 700, 1000, 1400, 2000\}$ (when reached) and the best checkpoint of every PHL and ModAL-1/3 run, we fix $M = 8$ minibatches of 400 molecules from each of Train, Test, IRC and NMS (the same minibatches for every run and estimator), compute the oracle gradient of the exact minibatch Hessian loss from all $3N$ columns, and draw $R = 16$ independent realisations of every estimator at the same parameters. The gate metric is the total variance of (9): $V_{\mathrm{total}} = V_{\mathrm{data}} + V_{\mathrm{cond}}$ with $V_{\mathrm{cond}} = \mathrm{mean}_{b,r} \|\hat{G}_{b,r} - G_b^\star\|^2$ and V_{data} the unbiased between-minibatch variance of $G_b^\star$ (summed over all parameters; no normalisation is needed for ratios at fixed parameters). Per dataset the gate statistic is the mean over audited checkpoints of the per-checkpoint ratio to PHL, with a 95% hierarchical bootstrap interval (runs, then minibatches and draws). A point estimate ≤ 2 meets the gate; an interval straddling 2 is flagged as borderline (so 2.04 ± 0.1 is a borderline failure and 1.96 ± 0.1 a borderline pass); per-stage ratios are reported in full, and the conditional ratio V_{cond} (estimator noise only) is reported as a secondary diagnostic.

Pilot data used for design. All design choices (probe law, score exponents, defensive weight, budgets) were made on a separate pilot seed (100) that does not enter any result: per-molecule gradient-noise diagnostics on 32–600 molecules per set at five checkpoints of a pilot PHL run, and a pilot audit. These runs and every abandoned alternative are documented in the attempts log.

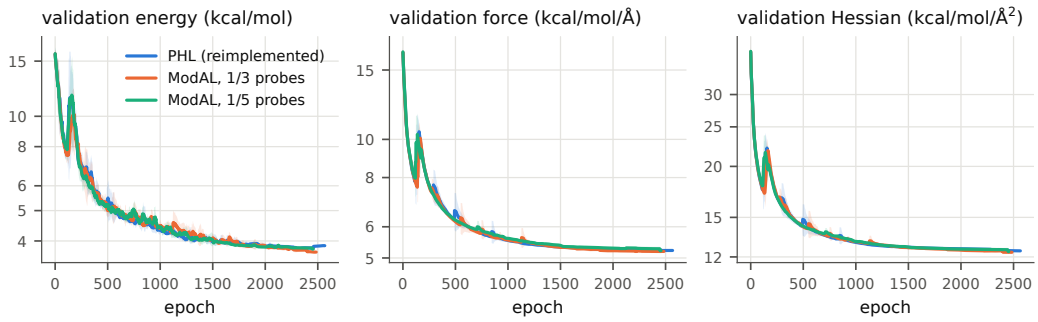


Figure 1: Validation RMSE during training (mean ± s.d. over five seeds of per-seed running medians over 31 epochs for energy and force and over 7 evaluations for the Hessian, which is evaluated exactly every 5 epochs). Curves end when the last seed stops.

5 Results

5.1 Probe count, accuracy and variance

Table 1 is the central result. MODAL with probe fraction $1/3$ evaluates 131.5 HVPs per SGD step on average (133 in every full minibatch), $3.01\times$ fewer than reimplemented PHL. Its Test/IRC/NMS RMSEs are statistically indistinguishable from reimplemented PHL in all nine cells under the pre-registered paired t-test ($9/9$ cells with $p > 0.05$; Table 2). The smallest p-value, 0.055 for the Test Hessian RMSE, is borderline; in that cell MODAL is the more accurate estimator (mean paired difference -0.7%). All mean paired differences lie between -1.9% and $+0.1\%$ of the PHL error (Table 3). Route (a) alone does not establish equivalence, so we also ran the secondary TOST analysis: it establishes equivalence within $\pm 10\%$ in $9/9$ cells. Averaged over training, its total gradient variance is 1.56 (Test), 1.76 (IRC) and 1.54 (NMS) times PHL's (95% CIs $[1.54, 1.61]$, $[1.73, 1.79]$, $[1.51, 1.58]$), so it meets the pre-registered $2\times$ gate on all three datasets. The fivefold variant (79.1 HVPs per step, $5.00\times$ fewer) passes the accuracy test in $9/9$ cells (TOST: $9/9$), but its gradient variance is 2.54/2.93/2.57 times PHL's, so it fails the variance gate on every dataset, as Proposition 4 predicts (Section 6.2). Across the eight Hessian-supervised configurations, 5 of 72 cell-level tests reject parity at $p \leq 0.05$ (about 3.6 would be expected by chance alone); none of them concerns either MODAL configuration, and the corresponding mean differences are at most 2.7% of the PHL error (Table 3). E–F training, as expected, differs from PHL in every cell.

5.2 Training dynamics and the variance trajectory

Validation curves of MODAL track those of PHL throughout training (Figure 1); runs stop by the same plateau rule after 1870–2486 epochs (PHL: 2272–2568). The variance ratio is not constant during training (Figure 2). On the evaluation sets it is highest in the first epochs and decreases as the model learns the curvature, which is the regime in which mode-aligned probes become most effective (Section 6.1); on the training set it rises again late in training (Section 6.3). Figure 3 shows the final RMSEs in the format of Fig. 3 of Rodriguez et al. [2026], next to the published PHL values, and Figure 4 the probe-count trace.

6 Analysis

6.1 Mechanism: what each ingredient buys

The noise is in the probes, and it is heavy tailed. Across all audited checkpoints and sets, the conditional (probe) variance of PHL's Hessian gradient is 2.2–5.4 times the between-minibatch variance of the exact gradient. Per-molecule diagnostics agree: probe/data ratios are 2.04–5.31 on the final runs and 2.08–4.24 on the pilot. Probe noise is therefore the larger target, but not the only one. The data term, which no subsampling scheme can reduce, is 16–31% of PHL's total variance, and it is multiplied by B/P for any estimator that probes P of B molecules. The exact per-molecule

Table 1: Main results. RMSEs are means $\pm$ standard deviations over five seeds (E: kcal/mol, F: kcal/mol/Å, H: kcal/mol/Å^2; exact Hessians; best-validation checkpoint). "Probes / minibatch step" is the time-averaged number of molecule-HVPs per SGD step over the whole run ($B = 400$; the last of the 71 minibatches has 69 molecules, hence 395.3 for PHL). Wall-clock: single-process training-phase time per epoch on one idle H200 (median of 20 epochs). Accuracy parity: two-sided paired t-test against reimplemented PHL (5 seed pairs), route (a); all nine cells are in Table 2. Gradient-variance ratio: total minibatch-gradient variance of the estimator divided by PHL's at the same parameters and minibatches, averaged over all audited checkpoints of the PHL and MODAL-$1/3$ runs, reported as Test/IRC/NMS (CIs in Table 4). Ablation rows share the seeds, data, minibatches and protocol of the first block. [a]Rodriguez et al. [2026], Fig. 3(a), p. 12 (randomised probes; bar values; no standard deviations are printed). [b]Ibid., Fig. 4, p. 16 (NVIDIA A6000, not comparable to our hardware).

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	Accuracy-parity test vs. reimpl. PHL (Test E)	Gradient-var ratio (your / reimpl. PHL)
PHL (paper as reported)[a]	B (400)	13.34[b]	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71	—	—
PHL (reimplemented, 5 seeds)	B (395.3)	1.83	3.87 ± 0.04	5.46 ± 0.03	12.80 ± 0.08	6.53 ± 0.08	6.98 ± 0.04	14.72 ± 0.12	12.86 ± 0.19	12.60 ± 0.15	36.23 ± 1.64	—	1.0
MODAL, $\phi = 1/3$ (this work)	131.5 (3.01× fewer)	2.39	3.80 ± 0.08	5.43 ± 0.08	12.71 ± 0.11	6.46 ± 0.12	6.93 ± 0.08	14.68 ± 0.11	12.85 ± 0.51	12.61 ± 0.34	35.76 ± 1.86	$p = 0.129$	1.56/1.76/1.54
MODAL, $\phi = 1/5$ (pre-registered limit probe)	79.1 (5.00× fewer)	2.45	3.88 ± 0.03	5.48 ± 0.02	12.74 ± 0.11	6.54 ± 0.14	6.98 ± 0.02	14.63 ± 0.03	12.78 ± 0.40	12.56 ± 0.20	35.24 ± 1.03	$p = 0.627$	2.54/2.93/2.57
Mode-aligned probes, all molecules	B (395.3)	1.83	3.87 ± 0.04	5.46 ± 0.03	12.77 ± 0.06	6.60 ± 0.04	6.99 ± 0.07	14.76 ± 0.05	12.95 ± 0.52	12.63 ± 0.34	33.58 ± 2.74	$p = 0.902$	0.75/0.74/0.63
Uniform HT, Gaussian, $\phi = 1/3$	131.5 (3.01× fewer)	2.40	3.82 ± 0.10	5.47 ± 0.08	12.70 ± 0.14	6.51 ± 0.19	6.95 ± 0.08	14.58 ± 0.12	12.72 ± 0.15	12.48 ± 0.11	34.76 ± 1.99	$p = 0.172$	3.13/3.06/3.12
Uniform HT, mode-aligned, $\phi = 1/3$	131.5 (3.01× fewer)	2.34	3.83 ± 0.06	5.43 ± 0.05	12.69 ± 0.12	6.52 ± 0.08	6.96 ± 0.05	14.63 ± 0.05	12.75 ± 0.20	12.57 ± 0.09	35.44 ± 0.96	$p = 0.268$	2.25/2.25/1.92
Curvature-weighted HT, Gaussian, $\phi = 1/3$	131.5 (3.01× fewer)	2.45	3.87 ± 0.01	5.44 ± 0.05	12.72 ± 0.12	6.52 ± 0.14	6.96 ± 0.04	14.68 ± 0.07	12.81 ± 0.15	12.48 ± 0.14	34.95 ± 1.07	$p = 0.836$	2.03/2.32/2.44
Uniform HT, Gaussian, $\phi = 1/5$	79.1 (5.00× fewer)	2.50	3.83 ± 0.06	5.47 ± 0.04	12.69 ± 0.09	6.59 ± 0.13	7.00 ± 0.04	14.59 ± 0.08	12.80 ± 0.43	12.60 ± 0.26	35.26 ± 1.39	$p = 0.182$	4.83/5.17/5.29
One-column probes, all molecules	B (395.3)	1.78	3.87 ± 0.05	5.47 ± 0.03	12.74 ± 0.05	6.58 ± 0.10	6.99 ± 0.03	14.70 ± 0.02	12.96 ± 0.16	12.60 ± 0.09	34.26 ± 0.91	$p = 0.978$	2.50/2.92/3.65
E–F (no Hessian term)	0	0.91	4.44 ± 0.04	4.66 ± 0.04	162.94 ± 3.08	7.31 ± 0.21	8.76 ± 0.33	161.45 ± 2.82	21.90 ± 0.66	28.66 ± 1.39	148.82 ± 5.91	$p = 4.2e-05$	—

Table 2: Paired two-sided t-test p-values against reimplemented PHL (5 seeds) for every configuration and cell. *: $p \leq 0.05$ (accuracy parity rejected under route (a)). †: TOST equivalence within $\pm 10\%$ of the PHL mean established ($p < 0.05$; secondary).

Configuration	Test E	Test F	Test H	IRC E	IRC F	IRC H	NMS E	NMS F	NMS H
MODAL, $\phi = 1/3$ (this work)	0.129†	0.387†	0.055†	0.344†	0.342†	0.326†	0.970†	0.929†	0.745†
MODAL, $\phi = 1/5$ (pre-registered limit probe)	0.627†	0.537†	0.266†	0.913†	0.987†	0.165†	0.678†	0.744†	0.216†
Mode-aligned probes, all molecules	0.902†	0.996†	0.540†	0.033*†	0.747†	0.519†	0.661†	0.863†	0.117
Uniform HT, Gaussian, $\phi = 1/3$	0.172†	0.918†	0.044*†	0.856†	0.586†	1.2e-04*†	0.180†	0.323†	0.237
Uniform HT, mode-aligned, $\phi = 1/3$	0.268†	0.204†	0.246†	0.885†	0.239†	0.044*†	0.328†	0.750†	0.281†
Curvature-weighted HT, Gaussian, $\phi = 1/3$	0.836†	0.439†	0.115†	0.914†	0.475†	0.438†	0.736†	0.296†	0.151†
Uniform HT, Gaussian, $\phi = 1/5$	0.182†	0.853†	0.102†	0.141†	0.473†	0.077†	0.813†	0.986†	0.046*†
One-column probes, all molecules	0.978†	0.688†	0.265†	0.182†	0.604†	0.717†	0.370†	0.971†	0.107
E–F (no Hessian term)	4.2e-05*	7.3e-06*	4.2e-08*	7.5e-04*	3.0e-04*	3.7e-08*	3.6e-06*	1.6e-05*	1.9e-06*

Table 3: Mean paired RMSE difference relative to reimplemented PHL, in percent (negative: lower error than PHL).

Configuration	Test E	Test F	Test H	IRC E	IRC F	IRC H	NMS E	NMS F	NMS H
MODAL, $\phi = 1/3$ (this work)	-1.9	-0.6	-0.7	-1.0	-0.6	-0.3	-0.1	+0.1	-1.3
MODAL, $\phi = 1/5$ (pre-registered limit probe)	+0.1	+0.3	-0.5	+0.1	+0.0	-0.6	-0.6	-0.3	-2.7
Mode-aligned probes, all molecules	-0.1	+0.0	-0.2	+1.0	+0.1	+0.3	+0.8	+0.2	-7.3
Uniform HT, Gaussian, $\phi = 1/3$	-1.3	+0.1	-0.8	-0.2	-0.4	-1.0	-1.0	-0.9	-4.1
Uniform HT, mode-aligned, $\phi = 1/3$	-1.2	-0.7	-0.9	-0.1	-0.3	-0.6	-0.8	-0.3	-2.2
Curvature-weighted HT, Gaussian, $\phi = 1/3$	-0.1	-0.4	-0.6	-0.1	-0.3	-0.3	-0.4	-1.0	-3.5
Uniform HT, Gaussian, $\phi = 1/5$	-1.1	+0.1	-0.9	+0.9	+0.4	-0.9	-0.4	+0.0	-2.7
One-column probes, all molecules	+0.0	+0.2	-0.5	+0.7	+0.2	-0.1	+0.8	-0.0	-5.4
E–F (no Hessian term)	+14.7	-14.7	+1172.7	+12.0	+25.6	+996.8	+70.3	+127.4	+310.8

gradients are nearly orthogonal, as Proposition 4 assumes: in the pilot diagnostics that store them, the squared norm of their mean is 0.8–17% of their mean squared norm (12 dataset×checkpoint cells, corrected for finite-sample bias), which shifts the predicted variance ratios by only a few percent because the data term is a small part of the total. Across molecules the one-probe second moments are heavy tailed: the top 5% of molecules carry 29–65% of the summed Gaussian second moment. s_m^2 is positively rank-correlated with the Hessian loss (0.80–0.89) and negatively with molecule size (-0.73 to -0.49), as the scaling $s_m^2 \propto L_m/n_m^2$ of Section 3 predicts.

Ablations separate the probe law from the allocation. Table 4 evaluates every estimator at the same parameters. With all molecules probed, mode-aligned Rademacher probes lower PHL's total variance to 0.75/0.74/0.63 of its value (Test/IRC/NMS). This is a zero-cost improvement of the probe law alone, and it is the control needed to separate the probe-law gain from the probe-count saving. Measured against this same-law, same-count baseline, MODAL-1/3 has 2.07–2.46 times the variance. Its threefold saving at the $2\times$ budget relative to Gaussian PHL is therefore partly paid for by the better law: relative to the mode-aligned baseline, and using the linear scaling in r confirmed in Section 6.2, the same budget would allow only about 2.4–2.9× fewer probes. At one third of the probes, uniform subsampling with Gaussian probes has the ratio 3.13/3.06/3.12, close to the value $B/P = 3$ that Proposition 4 gives for $\gamma_{\mathrm{law}} = \gamma_{\mathrm{alloc}} = 1$. Switching the law to mode-aligned probes lowers it to 2.25/2.25/1.92, switching the allocation to the curvature score lowers it to 2.03/2.32/2.44, and combining both (MODAL) gives 1.56/1.76/1.54. The law contributes a factor 1.37–1.62 and the allocation a factor 1.28–1.54, and the two compose approximately multiplicatively, as the factorisation $r/(\gamma_{\mathrm{law}}\gamma_{\mathrm{alloc}})$ implies. Neither ingredient alone keeps all three evaluation sets within the $2\times$ budget at a threefold reduction; the law alone does so only on NMS, and the allocation alone on none. One-column probing [Cui et al., 2026, Koker et al., 2026] has 2.50/2.92/3.65 times PHL's variance at the same probe count. At matched probe budget it is thus the noisiest law in the audit, in line with the pilot comparison of probe laws (Appendix, Table 8).

Why mode alignment works, and why it improves during training. By Proposition 2, the gain of a basis-Rademacher law over the Gaussian law equals the diagonal mass of the anticommutator matrices $\{E_m, D_{m,p}\}$ in that basis. In the pilot diagnostics, Rademacher probes in the eigenbasis of the current error (an oracle requiring $3N$ HVPs per molecule) reduce the per-probe gradient variance by 2.2–9.5×. The free reference normal-mode basis recovers a substantial part of this (1.2–4.4×,

Table 4: Gradient-variance audit. Ratio to PHL of the total minibatch-gradient variance (the gate; mean over audited checkpoints, 95% hierarchical-bootstrap CI) and of the conditional (probe-only) variance. Every estimator is evaluated at the same checkpoints (all audited checkpoints of the five PHL and five MODAL-$1/3$ runs), on the same $M = 8$ minibatches per set, with $R = 16$ draws.

	total-variance ratio (gate)			conditional ratio (secondary)		
Estimator	Test	IRC	NMS	Test	IRC	NMS
MODAL, $\phi = 1/3$ (this work)	1.56 [1.54, 1.61]	1.76 [1.73, 1.79]	1.54 [1.51, 1.58]	1.72	1.99	1.74
MODAL, $\phi = 1/5$ (pre-registered limit probe)	2.54 [2.49, 2.61]	2.93 [2.88, 3.00]	2.57 [2.52, 2.62]	2.96	3.51	3.15
Mode-aligned probes, all molecules	0.75 [0.74, 0.77]	0.74 [0.73, 0.75]	0.63 [0.62, 0.64]	0.69	0.66	0.48
Uniform HT, Gaussian, $\phi = 1/3$	3.13 [3.01, 3.29]	3.06 [3.00, 3.15]	3.12 [3.02, 3.24]	3.72	3.69	3.93
Uniform HT, mode-aligned, $\phi = 1/3$	2.25 [2.20, 2.31]	2.25 [2.21, 2.29]	1.92 [1.89, 1.96]	2.59	2.62	2.27
Curvature-weighted HT, Gaussian, $\phi = 1/3$	2.03 [1.99, 2.10]	2.32 [2.28, 2.37]	2.44 [2.39, 2.49]	2.32	2.72	2.98
Uniform HT, Gaussian, $\phi = 1/5$	4.83 [4.66, 5.06]	5.17 [5.01, 5.36]	5.29 [5.10, 5.53]	5.88	6.44	6.92
One-column probes, all molecules	2.50 [2.45, 2.57]	2.92 [2.86, 3.03]	3.65 [3.50, 3.82]	2.92	3.51	4.68



Figure 2: Total-variance ratio to PHL across training (mean over audited runs at each checkpoint, 95% bootstrap band). Horizontal lines: ratio 2 (gate) and 1.

median 1.66), whereas Cartesian Rademacher probes give a median of only 1.21 (Appendix, Table 8). Our interpretation is that a trained potential reproduces mode shapes better than mode curvatures. Its error is then close to spectral in the reference basis, and the amplitude noise that Gaussian probes pay along those modes is exactly what Rademacher signs remove. Early in training the error is unstructured and the gain is smaller; it grows as the model learns the curvature, which is why the variance ratios of the subsampling estimators with mode-aligned probes fall over training on the held-out sets (Figure 2; the training set is discussed in Section 6.3). The allocation gain also grows: the pilot's achieved allocation gain rises from 1.11–1.23 at epoch 25 to 1.15–1.45 later, as the remaining error concentrates in fewer molecules and the heavy tail that the score exploits gets heavier.

What the allocation does. The selection logs show that probes go where the variance theory says they should (Figure 5). Molecules with 4 atoms are probed 0.97 times per epoch on average and molecules with 23 atoms 0.17 times, against a uniform rate of $1/3$. The decile of highest reference curvature is probed 0.49 times per epoch and the lowest decile 0.23 times. Because of the defensive mixture no molecule is starved: the smallest inclusion probability in any run is 0.085, and every training molecule receives at least 0.114 probes per epoch on average, i.e. more than a hundred fresh probes over a run. We read this as one reason the reduced probe count does not cost accuracy. Unlike in the fixed-vector regime of Rodriguez et al. [2026], every molecule's curvature is still sampled in many independent directions over training.

6.2 The limit: why threefold and not fivefold

Proposition 4 turns per-molecule second moments into the largest reduction compatible with a variance budget. Figure 6 reports $r^\star(2)$ along training for four estimator classes, from per-molecule diagnostics on the pilot run (600 molecules per set) and on the final PHL run of seed 0 (300 molecules per set, noisier). Uniform subsampling with Gaussian probes is pinned at $r^\star = 2$. Mode-aligned

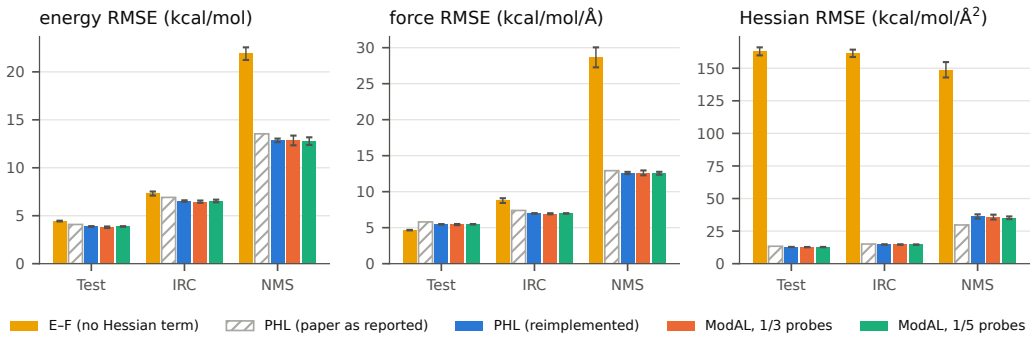


Figure 3: Final RMSEs (mean ± s.d. over five seeds) in the format of Fig. 3(a) of Rodriguez et al. [2026]; hatched bars are the published PHL values (no error bars are printed there).

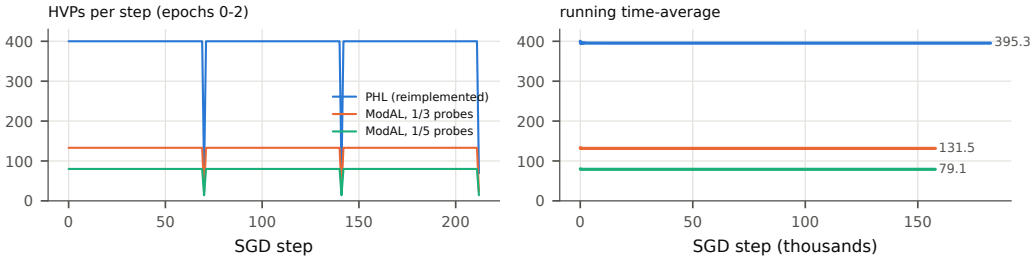


Figure 4: Probe count per SGD step for seed 0: the per-step count over the first three epochs (the dips are the final 69-molecule minibatch of each epoch) and its running time-average over the full run, whose final value is the contract metric.

probes with uniform allocation reach 2.38–3.48 (pilot). The zero-HVP score raises this to 2.84–4.17 (final run: 2.92–5.32). Even the oracle allocation that knows every s_m reaches 3.08–5.21 on the pilot, and exceeds fivefold there only on NMS at the last pilot checkpoint. On IRC, the practical limit stays between 2.84 and 3.28 at every pilot checkpoint. A fivefold reduction at variance parity would require $\gamma_{\text{law}}\gamma_{\text{alloc}} \geq 2.5$ on every dataset. The measured probe-law gains are 1.19–1.74 and the achieved allocation gains 1.11–1.45. The limit is dataset- and stage-dependent. Using the linear scaling in r confirmed below, the audited ratios from epoch 700 on imply variance-preserving reductions of about $4.3\times$ on Test and $4.2\times$ on NMS, but only $3.5\times$ on IRC and $3.2\times$ on the training set; averaged over all stages they are 3.8, 3.4 and $3.9\times$ on Test, IRC and NMS. A single probe budget must serve all of these sets and stages. The audit confirms the predicted scaling directly: at the same checkpoints, the variance ratio of the pre-registered fivefold variant is 1.62–1.67 times that of the threefold one on Test/IRC/NMS, against the factor $5/3 = 1.67$ that Proposition 4 implies when the γ factors are held fixed. The limit is a property of this estimator class, this data and this model, not of Hessian supervision in general. The error eigenbasis promises much larger law gains than the reference modes deliver, and both γ factors grew during training.

6.3 Failure analysis

Earliest training stage. At epoch 5, MODAL's variance ratio is slightly above two on every evaluation set (2.02–2.03; Table 6). We report this borderline stage-level failure as pre-registered; it is the only evaluation-set stage above two. On the training set the ratio is still 2.00 at epoch 25. Early in training the curvature error is large and unstructured, so the mode-aligned gain is at its smallest. The per-molecule noise is also still fairly homogeneous, so the score has little heavy tail to exploit (achieved pilot allocation gains 1.11–1.23 at epoch 25). This phase covers the first few dozen of more than two thousand epochs. A short warm-up with a larger probe fraction would remove it at negligible cost in the time-averaged count; we did not run that variant because it was not pre-registered.

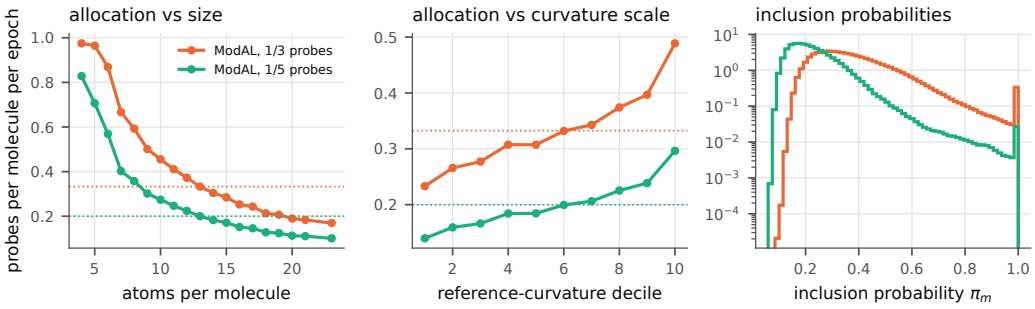


Figure 5: Where MODAL spends its probes (all seeds, from the per-step selection logs): probes per molecule per epoch against atom count and against the decile of reference curvature scale h_m (dotted: average rate), and the distribution of inclusion probabilities.

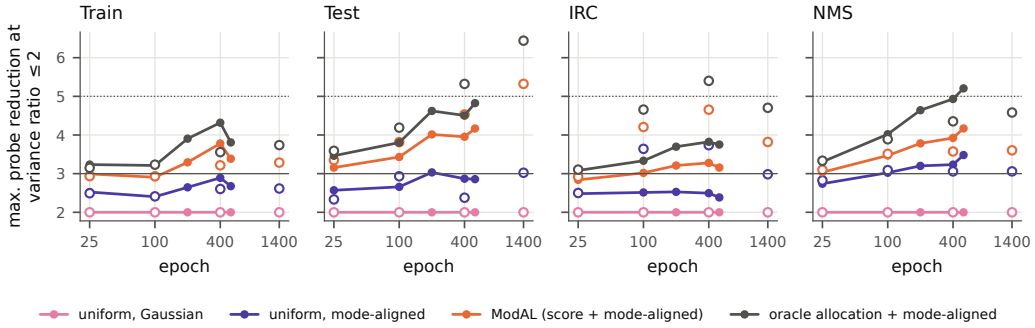


Figure 6: Largest probe reduction compatible with a total-variance ratio ≤ 2 (Proposition 4) from per-molecule diagnostics: pilot run (filled markers and lines, 600 molecules per set) and final PHL run of seed 0 (open markers, 300 molecules per set). Horizontal lines at $3\times$ and $5\times$.

IRC is the binding evaluation set; the training set is tighter still. Among the three evaluation sets, IRC has the highest time-averaged ratio (1.76, against 1.56 on Test and 1.54 on NMS). It also stays highest late in training: averaged over the checkpoints from epoch 700 on, it is 1.69, against 1.38 and 1.42. The ablations at the same checkpoints locate the deficit. The mode-aligned law helps IRC about as much as Test (uniform-allocation gains 1.37 and 1.39), but the allocation helps IRC less (gains 1.32 and 1.54). The per-molecule diagnostics give two reasons. First, there is less heterogeneity to exploit: the oracle allocation gain on IRC is 1.24–1.61 against 1.35–1.70 on Test. Second, the score ranks IRC molecules slightly less well: its Spearman correlation with s_m^2 is 0.80–0.82 on IRC against 0.81–0.87 on Test, and the drop comes from the force-residual term (0.49–0.66 against 0.58–0.72). Both reasons follow from what IRC geometries are: intermediate configurations along reaction paths, not stationary points. Their reference forces are large (RMS 5.75 kcal/mol/Å against 0.011 for RTP), their reference Hessians are no longer annihilated by rigid rotations (rotational leakage $13\times$ that of RTP), and 32% of them have a negative-curvature mode. A force residual is a clean signal of a poorly fitted region only when the reference force is near zero. NMS geometries are further from equilibrium still. There the force-residual signal is weakest (0.32–0.38) and so is the allocation gain (1.28). The mode-aligned law compensates: NMS has the largest law gain of all sets (1.62), presumably because its large distortions produce large errors in a few stiff modes that stay close to diagonal in the reference basis.

The training set, on which the estimator actually operates, is tighter than any evaluation set (1.90, CI $[1.88, 1.94]$). Its ratio rises again late in training, from 1.73 at epoch 200 to 1.89 on average from epoch 700 on, while the held-out ratios keep falling. Our interpretation is that curvature supervision removes the structured, mode-diagonal part of the error first on the molecules it is trained on, leaving a less structured residual for which mode alignment and allocation help less. The training-set ratio exceeds two at epoch 5 (2.09) and the best checkpoint (2.03), and stays at or below 2.00 at the other

checkpoints. The training set is not part of the pre-registered gate, but the margin in the regime that matters for optimisation is small, and at its last checkpoints it is essentially zero.

Fivefold reduction. The $\phi = 1/5$ variant fails the variance gate on all evaluation sets (2.54/2.93/2.57) and at every audited stage (lowest stage value 2.06), so we do not claim a fivefold variance-preserving reduction. Its accuracy is nevertheless close to PHL's (9 of 9 cells at parity under route (a); Table 2). At five seeds, route (a) of the accuracy test therefore cannot distinguish a threefold from a onefold increase in Hessian-gradient variance. More substantively, the downstream accuracy of PHL-type training is much less sensitive to Hessian-gradient variance than the variance gate is. One-column probing makes the same point from the other side: at full probe count it carries 2.50–3.65 times PHL's gradient variance, yet Rodriguez et al. [2026] report it as statistically indistinguishable from PHL in this regime, and so do our runs (9 of 9 cells at parity).

Transfer to the paper's convergence regime. Every run in this study trains to the published stopping rule, so there is no reduced schedule to extrapolate from. If anything, our runs train longer than the paper's: 2272–2568 epochs for PHL, against the ∼700–1000 reported. The notebook's learning rate of 10^{-3} needs seven halvings to reach the 10^{-5} stopping threshold, while the 10^{-4} stated in the supplementary information needs four. The reimplemented PHL is slightly more accurate than the published numbers in eight of nine cells (Table 1; e.g. Test energy 3.87 versus 4.09 kcal/mol, NMS energy 12.86 versus 13.54). The exception is the NMS Hessian RMSE, which is higher (36.23 versus 29.71 kcal/mol/Å^2). Candidate causes include the longer high-learning-rate phase, which fits curvature on near-equilibrium training geometries and may extrapolate it less well to distorted ones, the split permutation, and the evaluation of the published Hessian errors; we cannot separate them. All configurations share this protocol and these seeds, so the paired comparisons are unaffected. As for transfer, the variance mechanism strengthens as training proceeds, because the mode-aligned and allocation gains grow as the curvature error shrinks (Figure 2). A run that stopped at the paper's ∼1000 epochs would have spent a larger fraction of its steps in the high-ratio early phase. Even so, averaged over the checkpoints up to epoch 1000, the ratios are 1.65, 1.79 and 1.58 on Test, IRC and NMS, still below two.

6.4 Computational cost

MODAL's classical overhead per step is one score per molecule, computed from quantities already available, one random permutation and cumulative sum, and one $n_m \times n_m$ matrix–vector product per probed molecule. The normal-mode bases are computed once, by one dense eigendecomposition per molecule: 82 s for all 131,862 configurations of the three datasets on one GPU. They occupy as much memory as the stored reference Hessians. In wall-clock terms, however, the threefold probe saving does not pay off at this system size. A single-process epoch takes 2.39 s for MODAL-$1/3$, against 1.83 s for PHL and 0.91 s for E–F training. With 14-atom molecules in minibatches of 400, the step is limited by kernel-launch latency rather than arithmetic, and evaluating the HVPs on a separate graph for the selected molecules adds launches. The scaling benchmark of Figure 7 isolates this effect with non-interacting super-systems of 14–874 atoms. With 64 systems per step, MODAL-$1/3$ takes 1.41× PHL's step time at 13 atoms per system, breaks even between 216 and 434 atoms, and is 1.26× faster at 874 atoms (MODAL-$1/5$: 1.47×); at that size its curvature phase (step time minus E–F time) costs 0.67 of PHL's (0.49 for $1/5$), above the ideal $1/3$, partly because the selected molecules are forwarded a second time and partly because of per-step fixed costs. The probe count is therefore the hardware-independent figure of merit at the scale studied here, and wall-clock benefits require larger systems or a fused implementation. When reference HVPs are generated on demand by finite differences of quantum-chemical forces [Rodriguez et al., 2026], each avoided probe also avoids two force calculations, so the threefold reduction applies directly to the cost of the reference data.

7 Limitations and conclusion

Limitations. The study uses one architecture (ANI-style), one dataset family and molecules of 4–23 atoms, which is the regime of the published baseline. At this size the reduced probe count does not translate into a shorter epoch, and the benefit is only in HVP count (and in reference-HVP cost when reference curvature is generated on demand); the scaling benchmark puts the wall-clock break-even between 216 and 434 atoms per system, but we did not train on such systems. The mode-aligned law

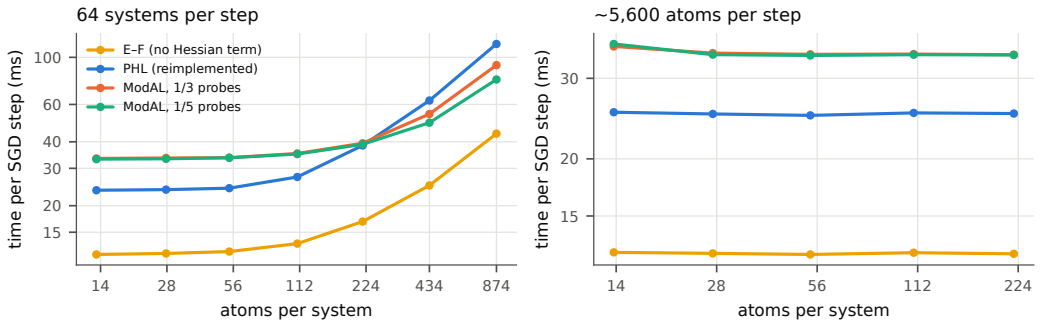


Figure 7: Time per training step (median over repetitions) against system size, using super-systems of k non-interacting RTP molecules placed 40 Å apart (block-diagonal Hessians and normal-mode bases), with 64 systems per step (left) or ≈5,600 atoms per step (right). Single process, idle H200.

needs eigenvectors of the reference Hessian. These are available whenever full reference Hessians are (the setting in which randomised PHL operates), but not when only reference HVPs are; unbiasedness holds for any orthogonal basis, so a basis from a cheaper Hessian model could be substituted, with unknown variance. Our accuracy gate is the paired t-test with five seeds, which does not by itself establish equivalence; the secondary TOST analysis does so at a $\pm 10\%$ margin, which is wide compared with the observed differences but narrower margins would need more seeds. The variance gate is met on average but not at the earliest audited checkpoint. Our reimplemented PHL is slightly more accurate than published in eight of nine cells but has a higher NMS Hessian error, and our runs train for more epochs than reported because the reference notebook's learning rate differs from the one stated in the paper's supplement; both differences affect all configurations alike. Finally, the limit of Proposition 4 is specific to single-probe Horvitz–Thompson estimators with zero-HVP allocation scores; estimators that reuse probes across steps, or learn error-adapted bases, are not covered by it.

Conclusion. Unbiased projected Hessian supervision does not need one probe per configuration per step. Most of the gradient noise in that regime is probe noise. Much of it is chi-square amplitude noise that Rademacher signs in the reference normal-mode basis remove at no cost, and the rest is concentrated in configurations that can be identified before any HVP is computed. MODAL uses this structure to cut HVPs per SGD step 3.01-fold on a reactive-chemistry benchmark, keeping accuracy statistically indistinguishable from PHL and gradient variance within the pre-registered budget. The same analysis explains why a fivefold cut does not preserve variance at this system size: a single-probe allocation can save at most $\kappa\gamma_{\mathrm{law}}\gamma_{\mathrm{alloc}}$, and the measured gains are too small. The analysis also says what would have to change for larger savings: a probe law closer to the error eigenbasis, which promises several-fold larger law gains than the reference modes deliver, or more heterogeneity across configurations for the allocation to exploit. How these two factors evolve with system size, where curvature supervision is most expensive, is the natural next question. Proposition 4 makes it answerable before any training run, from a few hundred per-molecule diagnostics.

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A Supplementary results

Per-seed results. Table 5 lists every seed of the reimplemented PHL and of both MODAL configurations; all other configurations are in `logs/results_summary.json`.

Table 5: Per-seed final RMSEs (best-validation checkpoint; E: kcal/mol, F: kcal/mol/Å, H: kcal/mol/Å^2), training length and time-averaged probes per step.

Config.	seed	epochs	probes/step	Test E	Test F	Test H	IRC E	IRC F	IRC H	NMS E	NMS F	NMS H
PHL	0	2568	395.3	3.90	5.44	12.73	6.52	6.96	14.62	12.88	12.59	33.68
PHL	1	2458	395.3	3.82	5.44	12.75	6.63	6.96	14.57	12.81	12.46	36.51
PHL	2	2346	395.3	3.93	5.48	12.94	6.54	6.94	14.77	13.08	12.79	38.18
PHL	3	2272	395.3	3.86	5.51	12.82	6.41	7.04	14.78	12.93	12.72	36.00
PHL	4	2289	395.3	3.85	5.45	12.78	6.55	6.99	14.85	12.58	12.44	36.78
MODAL 1/3	0	2223	131.5	3.69	5.41	12.68	6.42	6.88	14.67	13.70	13.10	38.13
MODAL 1/3	1	2486	131.5	3.79	5.37	12.59	6.42	6.86	14.49	12.39	12.18	33.53
MODAL 1/3	2	1870	131.5	3.87	5.57	12.87	6.68	7.06	14.80	12.87	12.62	36.43
MODAL 1/3	3	2398	131.5	3.76	5.39	12.63	6.41	6.93	14.74	12.73	12.69	36.44
MODAL 1/3	4	2211	131.5	3.88	5.42	12.78	6.39	6.94	14.70	12.55	12.47	34.25
MODAL 1/5	0	2222	79.1	3.91	5.51	12.71	6.74	7.02	14.60	13.35	12.69	34.89
MODAL 1/5	1	2123	79.1	3.84	5.46	12.82	6.39	6.98	14.63	12.98	12.79	36.38
MODAL 1/5	2	2063	79.1	3.91	5.48	12.88	6.58	6.96	14.67	12.69	12.54	36.30
MODAL 1/5	3	2458	79.1	3.87	5.45	12.59	6.44	6.98	14.65	12.51	12.50	34.38
MODAL 1/5	4	2446	79.1	3.85	5.50	12.70	6.55	6.96	14.61	12.36	12.27	34.27

Table 6: Total-variance ratio to PHL by checkpoint (mean over audited PHL and MODAL-$1/3$ runs).

	MODAL, $\phi = 1/3$				MODAL, $\phi = 1/5$			
Checkpoint	Train	Test	IRC	NMS	Train	Test	IRC	NMS
5 (10)	2.09	2.03	2.03	2.02	3.43	3.42	3.36	3.35
25 (10)	2.00	1.89	1.97	1.77	3.24	3.14	3.28	3.03
100 (10)	1.88	1.63	1.78	1.54	3.01	2.67	2.97	2.50
200 (10)	1.73	1.66	1.68	1.56	2.79	2.67	2.81	2.66
400 (10)	1.75	1.53	1.70	1.42	2.78	2.47	2.74	2.34
700 (10)	1.86	1.41	1.69	1.39	3.07	2.28	2.79	2.32
1000 (10)	1.91	1.43	1.70	1.40	3.14	2.28	2.89	2.38
1400 (10)	1.87	1.39	1.65	1.43	3.03	2.19	2.78	2.35
2000 (9)	1.94	1.31	1.74	1.46	3.29	2.06	2.80	2.40
best (10)	2.03	1.34	1.63	1.42	3.25	2.15	2.85	2.32

Variance ratio by training stage. Table 6 gives the total-variance ratio to PHL at every audited checkpoint, averaged over the audited runs (number of runs in parentheses; later checkpoints exist only for runs that trained that long). These are the values plotted in Figure 2.

Empirical check of unbiasedness. Proposition 1 is exact; as a check of the implementation we drew 256 independent realisations of each estimator on one fixed minibatch per evaluation set, at the epoch-100 checkpoint of PHL seed 0, and compared the means with the exact values from all $3N$ columns (Table 7). All relative biases of the loss are within $\pm 0.7\%$ and all 95% intervals contain zero. The squared norm of the mean gradient error, divided by its expectation under zero bias (V_{cond}/R), lies between 0.65 and 1.29, as expected for an unbiased estimator.

Table 7: Mean of R estimator draws against the exact minibatch Hessian loss and gradient (code: `code/unbiasedness_check.py`).

Set	Estimator	exact $\mathcal{L}_H$	mean of 256 draws	rel. bias [95% CI]	$\frac{\|\bar{G}-G^\star\|^2}{V_{\mathrm{cond}}/R}$
Test	PHL	9.3024e-04	9.3514e-04	+0.53% [-0.06, +1.12]	1.29
Test	MODAL 1/3	9.3024e-04	9.3580e-04	+0.60% [-0.24, +1.44]	1.16
Test	MODAL 1/5	9.3024e-04	9.2487e-04	-0.58% [-1.68, +0.53]	0.78
IRC	PHL	9.7258e-04	9.7361e-04	+0.11% [-0.42, +0.64]	0.96
IRC	MODAL 1/3	9.7258e-04	9.6604e-04	-0.67% [-1.35, +0.01]	1.13
IRC	MODAL 1/5	9.7258e-04	9.7349e-04	+0.09% [-0.90, +1.09]	0.82
NMS	PHL	4.0431e-03	4.0342e-03	-0.22% [-0.80, +0.36]	1.07
NMS	MODAL 1/3	4.0431e-03	4.0389e-03	-0.10% [-0.78, +0.57]	0.96
NMS	MODAL 1/5	4.0431e-03	4.0312e-03	-0.30% [-1.25, +0.66]	0.65

B Design diagnostics (pilot seed, not used for evaluation)

All design decisions were taken on pilot seed 100. Table 8 summarises the per-probe gradient-variance gain over Gaussian probes of every probe law we tried, measured on 32–64 molecules per set at pilot checkpoints (epochs 25–200). With 16–32 draws per molecule these estimates are noisy; values below one for Rademacher laws are Monte-Carlo error, since Proposition 2 guarantees a gain of at least one in expectation. One-column probes, which concentrate the probe on a single coordinate, were 2.1–4.2 times noisier than Gaussian probes in every cell; eigen-direction importance probes, which concentrate it on a single normal mode, were 1.7–2.4 times noisier in 7 of 8 cells. Rademacher probes in the eigenbasis of the current model Hessian, or of the current error (an oracle), need $3N$ extra HVPs per molecule and were used only to bound what basis choice can achieve. The allocation score exponents and the defensive weight were selected by the worst-case predicted variance ratio over Test/IRC/NMS and five pilot checkpoints (attempts log, entry A5); the chosen values are within 0.3% of the best grid point. Control variates built from per-molecule force- and energy-loss gradients were rejected because their mean cosine similarity with the exact Hessian-loss gradient is at most 0.13 in absolute value.

Table 8: Per-probe gradient-variance gain over Gaussian probes (pilot diagnostics; median and range over dataset×checkpoint cells). ‡Requires $3N$ extra HVPs per molecule (diagnostic only).

Probe law	median gain	range	#cells	needs extra HVPs?
Cartesian Rademacher	1.21	0.83–3.69	12	no
sphere (normalised Gaussian)	1.09	0.81–1.91	8	no
one-column (one-hot)	0.29	0.24–0.47	8	no
eigen-direction importance	0.48	0.42–1.86	8	no
Rademacher, model-Hessian basis‡	1.65	0.81–3.75	4	yes ($3N$)
Rademacher, error basis (oracle)‡	2.67	2.23–9.46	4	yes ($3N$)
Rademacher, reference normal modes	1.66	1.19–4.42	12	no

C Implementation and reproducibility

The code reuses the authors' architecture, optimisers, scheduler, loss normalisations and reverse-over-reverse HVP construction verbatim and re-implements only the data path, which in the authors' repository requires patching the installed TorchANI package (see `codebase_notes.md`). Every run logs its configuration, the training-set order (from which every minibatch is determined), the per-step probe count, the per-step selections and inclusion probabilities (MODAL and the subsampling ablations), the probe vectors of epoch 0, per-epoch validation metrics and wall-clock times, periodic checkpoints, and per-molecule errors of the final evaluation. Probes and selections of step t of seed s are drawn from a generator seeded with $10{,}000{,}019\,s + t$, so they can be regenerated exactly, and runs interrupted for scheduling resume from saved optimiser and scheduler states without changing their random draws. The pre-registration document, the attempts log, the variance-audit records (minibatch indices, per-draw squared errors and oracle norms for every estimator) and all scripts that generate the tables and figures of this paper are included with the logs.